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COMPUTER NETWORKS

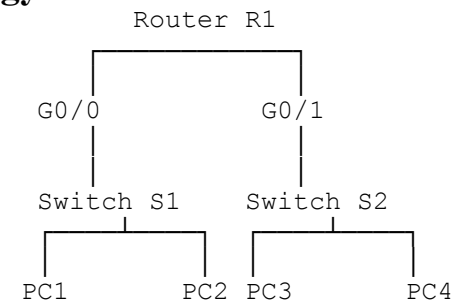
Ex.No.10	Design a simple topology and configure with one router, two switches and PCs using Cisco Packet Tracer
DATE:	

AIM:

To design and configure a simple network topology with one router, two switches, and four PCs in Cisco Packet Tracer for successful inter-network communication.

PROCEDURE:

1. Topology



2. Devices to place

In **Cisco Packet Tracer**, place:

- **1 Router** – Cisco 1941
- **2 Switches** – Cisco 2960
- **4 PCs**

Use **Copper Straight-Through** cables.

3. IP Addressing

Device	Interface	IP Address	Subnet Mask	Gateway
R1	G0/0	192.168.1.1	255.255.255.0	—
R1	G0/1	192.168.2.1	255.255.255.0	—
PC1	NIC	192.168.1.10	255.255.255.0	192.168.1.1
PC2	NIC	192.168.1.11	255.255.255.0	192.168.1.1

Device	Interface	IP Address	Subnet Mask	Gateway
PC3	NIC	192.168.2.10	255.255.255.0	192.168.2.1
PC4	NIC	192.168.2.11	255.255.255.0	192.168.2.1

So we have **two different networks**:

```
Network 1: 192.168.1.0/24
       PC1 ─┐
       PC2 ─┤
            S1 ── R1 G0/0 = 192.168.1.1

Network 2: 192.168.2.0/24
       PC3 ─┐
       PC4 ─┤
            S2 ── R1 G0/1 = 192.168.2.1
```

4. Configure Router R1

Click **R1 → CLI** and enter:

```
enable
configure terminal

interface gigabitEthernet 0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
exit

interface gigabitEthernet 0/1
ip address 192.168.2.1 255.255.255.0
no shutdown
exit

end
write memory
```

The important command is:

```
no shutdown
```

It activates the router interface.

Check the interfaces using:

```
show ip interface brief
```

You should see the interfaces as **up/up**.

SCREENSHOTS:(to be pasted)

5. Configure PC1

Go to:

PC1 → Desktop → IP Configuration

Enter:

```
IP Address:       192.168.1.10
Subnet Mask:      255.255.255.0
Default Gateway:  192.168.1.1
```

SCREENSHOTS:(to be pasted)

6. Configure PC2

```
IP Address:       192.168.1.11
Subnet Mask:      255.255.255.0
Default Gateway:  192.168.1.1
```

SCREENSHOTS:(to be pasted)

7. Configure PC3

```
IP Address:       192.168.2.10
Subnet Mask:      255.255.255.0
Default Gateway:  192.168.2.1
```

SCREENSHOTS:(to be pasted)

8. Configure PC4

```
IP Address:       192.168.2.11
Subnet Mask:      255.255.255.0
Default Gateway:  192.168.2.1
```

SCREENSHOTS:(to be pasted)

9. Test the network

From **PC1 → Desktop → Command Prompt**, type:

```
ping 192.168.1.11
```

This tests PC1 → PC2.

Then:

```
ping 192.168.2.10
```

This tests **PC1 → Router → Switch 2 → PC3**.

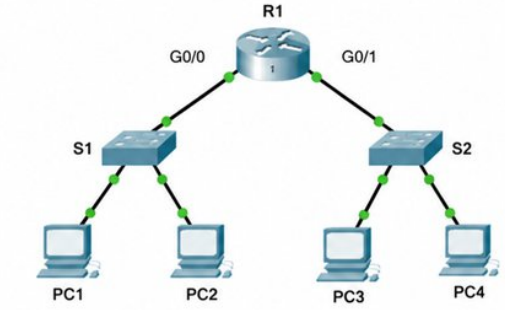
You should receive:

```
Reply from 192.168.2.10
Reply from 192.168.2.10
Reply from 192.168.2.10
Reply from 192.168.2.10
```

SCREENSHOTS:(to be pasted)

SAMPLE CONFIGURATION:

TOPOLOGY: One Router, Two Switches and PCs



IP ADDRESSING TABLE

Device	Interface	IP Address	Subnet Mask	Default Gateway
R1	G0/0	192.168.1.1	255.255.255.0	-
R1	G0/1	192.168.2.1	255.255.255.0	-
PC1	NIC	192.168.1.10	255.255.255.0	192.168.1.1
PC2	NIC	192.168.1.11	255.255.255.0	192.168.1.1
PC3	NIC	192.168.2.10	255.255.255.0	192.168.2.1
PC4	NIC	192.168.2.11	255.168.255.0	192.168.2.1

CONFIGURATION STEPS

1. Place Devices

- 1 Router (e.g., 1941)
- 2 Switches (e.g., 2960)
- 4 PCs
- Connections: Copper Straight-Through

2. Configure Router (R1)

```
R1> enable
R1# configure terminal
R1(config)# interface g0/0
R1(config-if)# ip address 192.168.1.1 255.255.255.0
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)# interface g0/1
R1(config-if)# ip address 192.168.2.1 255.255.255.0
R1(config-if)# no shutdown
R1(config-if)# end
R1# write memory
```

3. Configure PCs

For PC1

- IP Address: 192.168.1.10
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.1.1

For PC2

- IP Address: 192.168.1.11
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.1.1

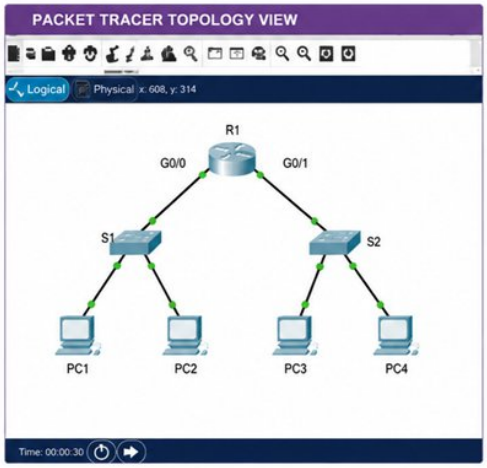
For PC3

- IP Address: 192.168.2.10
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.2.1

For PC4

- IP Address: 192.168.2.11
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.2.1

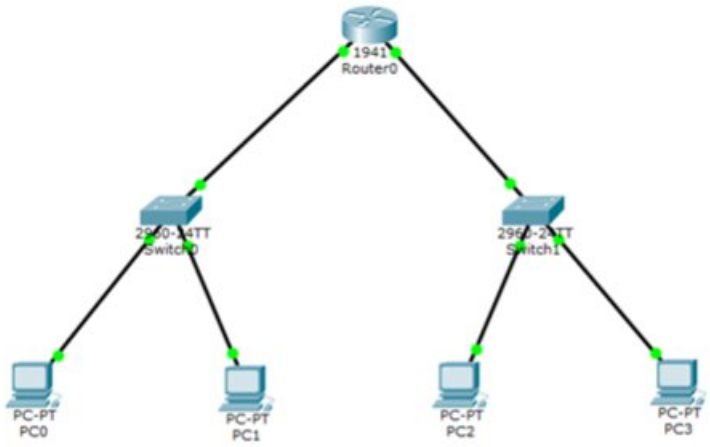
PACKET TRACER TOPOLOGY VIEW

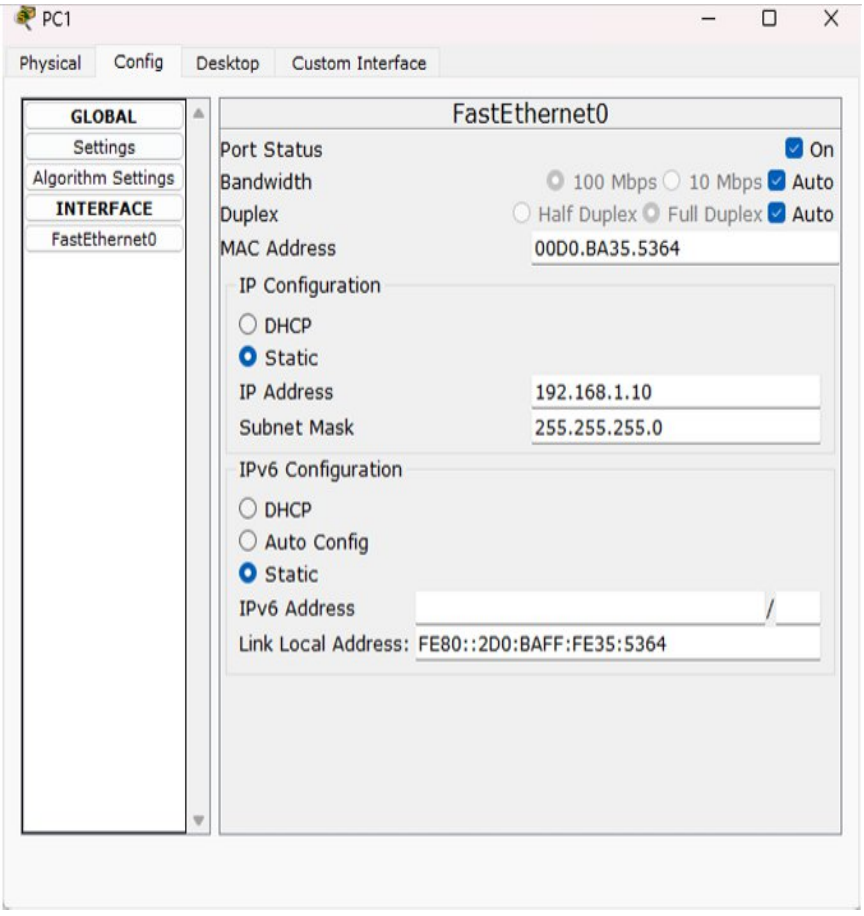


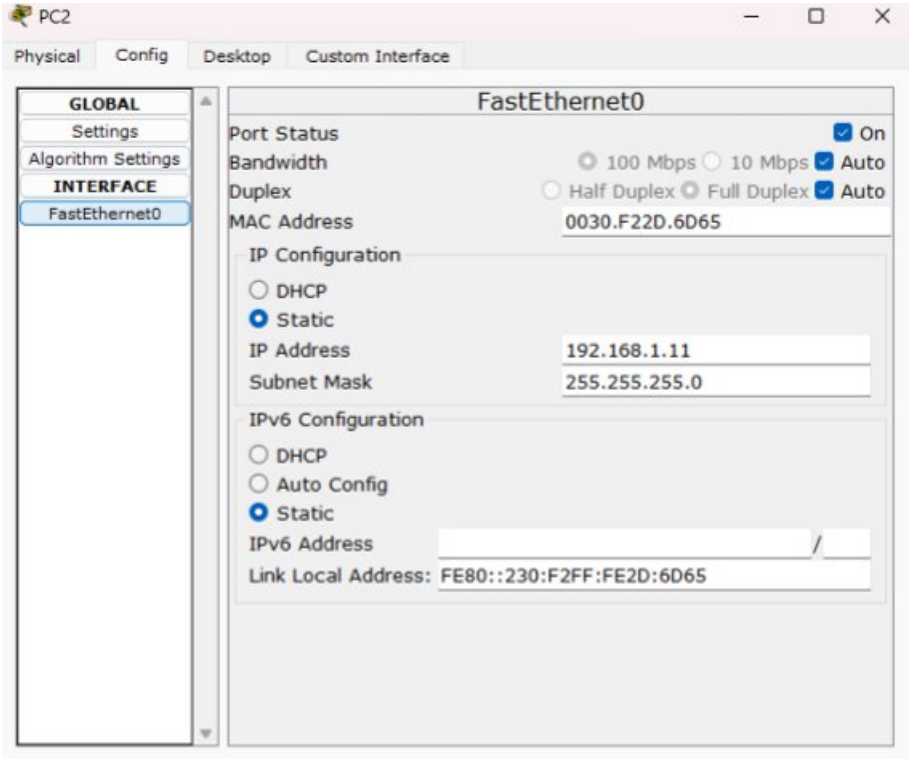
VERIFY CONNECTIVITY

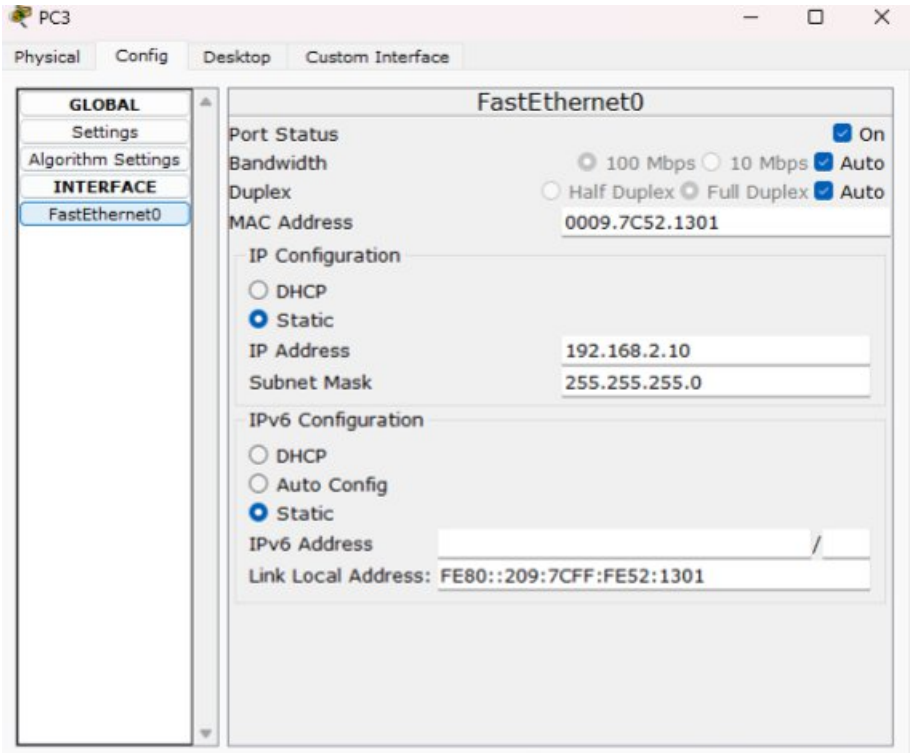
From any PC, open Command Prompt and ping other PCs in the same network and across networks.
Example: From PC1 ping PC3 → `ping 192.168.2.10`
All pings should be successful.

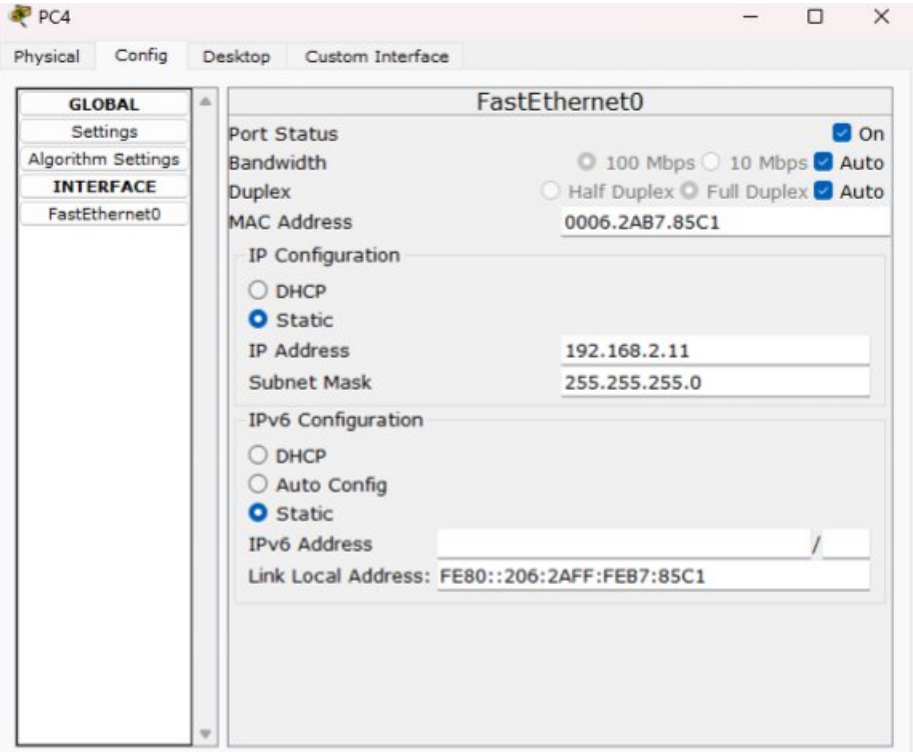
OUTPUT:

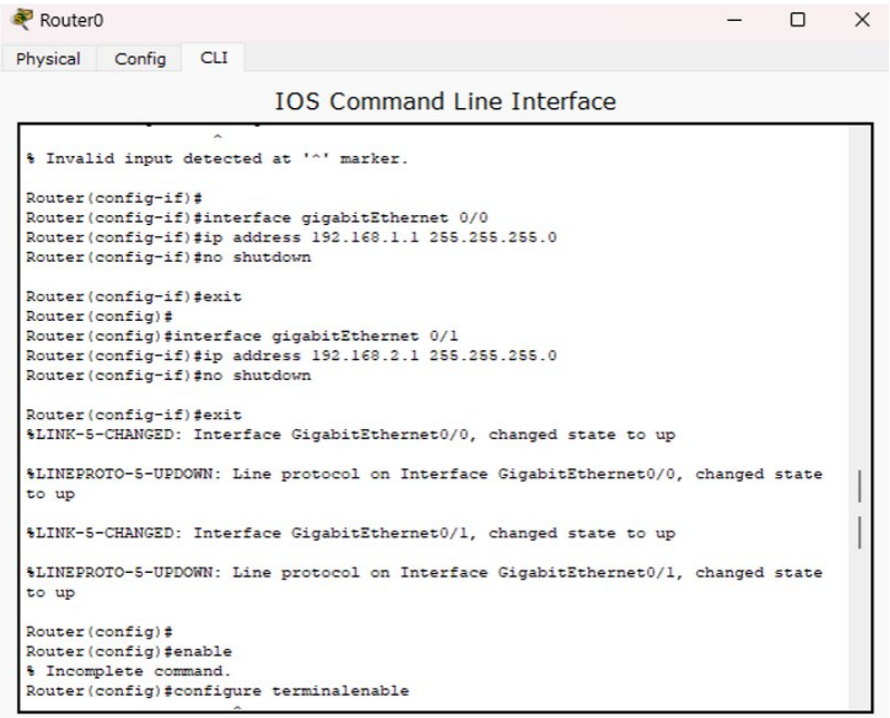

1941
Router0
PC-PT
PC0
PC-PT
PC1
PC-PT
PC2
PC-PT
PC3


PC1
Physical
Config
Desktop
Custom Interface
GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
FastEthernet0
Port Status
On
Bandwidth
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
00D0.BA35.5364
IP Configuration
DHCP
Static
IP Address
192.168.1.10
Subnet Mask
255.255.255.0
IPv6 Configuration
DHCP
Auto Config
Static
IPv6 Address
Link Local Address:
FE80::2D0:BAFF:FE35:5364


PC2
Physical
Config
Desktop
Custom Interface
GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
FastEthernet0
Port Status
On
Bandwidth
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
0030.F22D.6D65
IP Configuration
DHCP
Static
IP Address
192.168.1.11
Subnet Mask
255.255.255.0
IPv6 Configuration
DHCP
Auto Config
Static
IPv6 Address
Link Local Address:
FE80::230:F2FF:FE2D:6D65


PC3
Physical
Config
Desktop
Custom Interface
GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
FastEthernet0
Port Status
On
Bandwidth
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
0009.7C52.1301
IP Configuration
DHCP
Static
IP Address
192.168.2.10
Subnet Mask
255.255.255.0
IPv6 Configuration
DHCP
Auto Config
Static
IPv6 Address
Link Local Address:
FE80::209:7CFF:FE52:1301





RESULT:

A simple network topology with one router, two switches, and PCs was successfully designed and configured using Cisco Packet Tracer.Connectivity between the network devices and PCs was successfully established and verified.